

0D Field Theory

Shravan S. N.

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1 The Cumulant Expansion

We aim to find a perturbative series for the energy using a 0D path integral, and we'll treat it generally, i.e. without assuming the system is that of an anharmonic oscillator. We start by defining the partition function for the unperturbed system:

$$Z_0 = \int e^{-A_0} d\phi \quad (1)$$

where A_0 is some function of ϕ that represents the unperturbed Hamiltonian (in our case we might think of it as $\propto \phi^2$). We can then take inspiration from Boltzmann statistics to define the average of some observable as follows:

$$\langle f(\phi) \rangle = \frac{1}{Z_0} \int e^{-A_0} f(\phi) d\phi \quad (2)$$

Now, when we add the perturbation for the total partition function, Z , we can naturally write it using (2) as follows:

$$Z = \int e^{-A_0 - A_1} d\phi \quad (3)$$

$$= Z_0 \langle e^{-A_1} \rangle \quad (4)$$

where we've just separated the perturbation and used (2). The primary object of our interest will now be the average term in (4). Using the Taylor expansion of the exponential function and the linearity of expectation values, we find:

$$\langle e^{-A_1} \rangle = \left\langle 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} A_1^n \right\rangle = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \langle A_1^n \rangle \quad (5)$$

To find the energy we normally take the logarithm of the partition function, and so this is what we'll do. Again, we will use the Taylor expansion for $\ln(1+x)$ as we've suggestively written already in (5), which yields:

$$\begin{aligned} \ln \langle e^{-A_1} \rangle &= \sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{m} \left(\sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \langle A_1^n \rangle \right)^m \\ &= \sum_{m=1}^{\infty} \sum_{n_1=1}^{\infty} \cdots \sum_{n_m=1}^{\infty} (-1)^{m-1} (-1)^{n_1+\cdots+n_m} \frac{1}{m} \frac{\langle A_1^{n_1} \rangle \cdots \langle A_1^{n_m} \rangle}{n_1! \cdots n_m!} \end{aligned} \quad (6)$$

Now, ultimately we treat A_1 as small, so the point of this expansion is to treat it order by order. It is clear the order of each term in (6) is $n_1 + n_2 + \dots + n_m$, and so the object of interest is to group up all the terms for which this sum is equal. It is clear that this is precisely what the partition function counts with one small twist: permutation matters. When the permutation matters apparently this is called “compositions” instead of “partitions”, and the set of all compositions is creatively denoted $C(n)$. So, for instance:

$$C(3) = \{\{3\}, \{2, 1\}, \{1, 2\}, \{1, 1, 1\}\} \quad (7)$$

We will use (7) such that the length of each set inside $C(n)$ will be m from (6), and the numbers within the set will define $n_1, \dots, n_m$, which is why permutation matters. The main index will be the order of the term which we’ve already found to be $n_1 + \dots + n_m$, and as we’ve mentioned $m = |c|$ where $c \in C(n)$ is the particular composition we’re using, and finally the i -th number in the composition defines n_i . It immediately follows that we can write (6) as:

$$\ln \langle e^{-A_1} \rangle = \sum_{n=1}^{\infty} (-1)^n \sum_{c \in C(n)} \frac{(-1)^{|c|-1}}{|c|} \prod_{c_i \in c} \frac{\langle A_1^{c_i} \rangle}{c_i!} \quad (8)$$

This allows us to write the energy using (4) as:

$$E_0 = -\ln(Z) \quad (9)$$

$$\begin{aligned} &= -\ln Z_0 - \ln \langle e^{-A_1} \rangle \\ &= -\ln Z_0 + \langle A_1 \rangle - \frac{1}{2} (\langle A_1^2 \rangle - \langle A_1 \rangle^2) + \dots \end{aligned} \quad (10)$$

2 Diagrams

Though we haven’t shown it rigorously, the expression (10) is precisely the linked cluster theorem for 0D field theory; this is more apparent if you think of $\langle A_1^n \rangle$ as the sum of all n -th order diagrams (which haven’t taken into account indistinguishability). The general n -th order term just ends up being the sum of connected n -th order diagrams as the linked cluster theorem stipulates.

This leads us to our next endeavor, which is to come up with a diagrammatic “dictionary” to represent the sum (10) using a sum of linked diagrams, and we will do this for a concrete system, taking $A_0 = a\phi^2$, $A_1 = b\phi^4$ so that the partition function is:

$$Z = \int e^{-a\phi^2 - b\phi^4} d\phi \quad \longrightarrow \quad Z_0 = \int e^{-a\phi^2} d\phi \quad (11)$$

Now, using (12) we can write (10) and for our particular system as:

$$E_0 = E_0^{(0)} + \frac{3b}{4a^2} - \frac{3b^2}{a^4} + \dots \quad (12)$$

This is enough to get started on a diagrammatic approach. Let’s first find the value of a free propagator, i.e. a line in the diagram. This is given by $\langle \phi^2 \rangle$, with the average

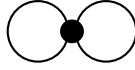
defined in (2). This integral, and every integral needed to compute (10) is trivial using the result:

$$\int_{-\infty}^{\infty} x^{2n} e^{-ax^2} dx = \frac{(2n-1)!!}{(2a)^n} \sqrt{\frac{\pi}{a}} \quad (13)$$

and in this case we see that the propagator is simply:

$$G = \langle \phi^2 \rangle = \frac{1}{2a} \quad (14)$$

The vertices on the diagram are ordinarily elements of the perturbing potential, and since the potential is just one number it's reasonable to guess each vertex simply picks up a factor of b . Now, when we try to draw out all the linked diagrams at each order, we see that the only distinct diagrams at first and second order are:



(i) First Order



(ii) Second Order

By distinct we are referring to the plausible edge connections, so for instance at second order there were only two possibilities: every line from one vertex goes to the other, or two lines go from one vertex to another with a self-loop at both. It's obviously not possible for only one (or three) line(s) to go between the two, and if no lines go between the two it wouldn't be a linked diagram so that's forbidden. Therefore, those are the only two distinct options.

We see that at first order there are two propagator lines and one dot, whereas at second order there are four propagator lines and two dots. If we combine our prescriptions of the dots and propagators (i.e. dots are b and lines are $1/(2a)$), we find that, as is, our first order prediction with the diagrams is proportional to $b/(2a)^2$, whereas the second order prediction is proportional to $b^2/(2a)^4$. This motivates us to write (12) in a more suggestive manner:

$$E_0 = E_0^{(0)} + \frac{3b}{(2a)^2} - \frac{48b^2}{(2a)^4} + \dots \quad (15)$$

The present goal is then to figure out where the combinatorial factors in (15) at each order are coming from.

We do this by first thinking of how we could generate every possible diagram, which all have a general shape as we've discussed. One way to think of it is by taking the general shape, and then permuting every vertex's legs (thinking of the legs as distinct, say you put a label on each of them), and each vertex has $4!$ of these permutations, for a total combinatorial factor of $(4!)^n$ for an n -th order diagram. However, this grossly overcounts the diagrams, for three reasons:

1. With self-loops, swapping the legs involved does not generate a new diagram. So every self-loop overcounts doubly.
2. Lines that go from one vertex to another can be interchanged, in the sense that line 1 and line 2 have their beginnings swapped, and their ends swapped. This does not yield a unique pairing, and hence this is an overcounting factor. If there are m lines from one vertex to another, we see that there are $m!$ ways to order the legs on both sides to yield the same pairings.
3. The dots are not distinct. That is, swapping the dots does not produce a new diagram. This is because we are only interested in distinct connections, not any particular vertex.

The first reason corresponds to a factor of 2 for every self-loop, the second reason corresponds to a factor of $m!$ for m “equivalent” lines, and the third reason corresponds to a factor of $1/n!$ for an n -th order diagram.

So, let’s take first order as an example. There is one dot, two self loops, and the loops are “equivalent” lines and so the combinatorial factor is:

$$\frac{(4!)^1}{(2)^2 \times (2)^1} = 3 \quad (16)$$

which is indeed correct. Moving onto second order, we find for the diagram on the left there are two dots, four equivalent lines. For the second diagram there are again two dots, two equivalent lines, and two self loops, and so the predicted combinatorial factor is:

$$\frac{(4!)^2}{4! \times 2!} + \frac{(4!)^2}{(2)^2 \times 2! \times 2!} = 12 + 36 = 48 \quad (17)$$

and we see this is perfectly in alignment with (15). To accommodate the flipping negative signs and the combinatorial factor of $(4!)^n$ we simply assign to the vertices a factor of $(-4! \times b)$ instead of (b) .

This, then, gives us our diagram dictionary in Figure 1 (see next page). The unfilled vertices just exist for clarity’s sake, but they are not part of the diagram. And for the last entry, the two vertices could very well be the same, and the value nevertheless holds.

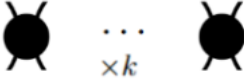
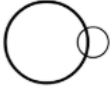
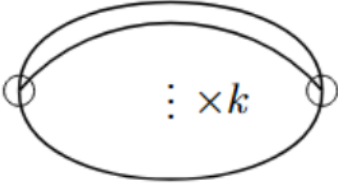
Diagram	Value
 k vertices	$\frac{(4! \times (-b))^k}{k!}$
 free propagator	$\frac{1}{2a}$
 self-loop	$\frac{1}{2}$
 k equivalent lines	$\frac{1}{k!}$

Figure 1: Diagram Dictionary